examples of pentavalent impurities are Arsenic (Atomic No. 33) and Antimony (Atomic No.51). Such impurities that produce n-type semiconductor are known as donor impurities, because they donate or provide free electrons to the semiconductor crystals.

To explain the formation of n-type semiconductor, consider a pure germanium crystal. We know that germanium atom has four valence electrons. When small amount of pentavalent impurity, like Arsenic is added to Germanium crystal, large number of free electrons available in the crystal. Arsenic is pentavalent i.e., its atom has five valence electrons. An Arsenic atom fits in the Germanium crystal in such a way that its four valence electron form covalent bonds with four Germanium atoms. The fifth valence electron of Arsenic atom finds no place in covalent bond and is thus become free electron as shown in Figure3.22. Therefore, for each Arsenic atom added, one free electron will be available in the Germanium crystal. Hence, an extremely small amount of Arsenic impurity provides enough atoms to supply millions of free electrons.

Figure 3.22 shows the energy band description of n-type semiconductor. The addition of pentavalent impurity has produced a number of conduction band electrons, i.e., free electrons. Therefore, valence electrons of pentavalent atom form covalent bonds with four neighbouring Germanium atoms. The fifth left-over valence electron of the pentavalent atom cannot be accommodated in the valence band; hence travels to the conduction band. The current flow in n-type semiconductor is shown in Figure 3.23. The following points may be noted carefully:

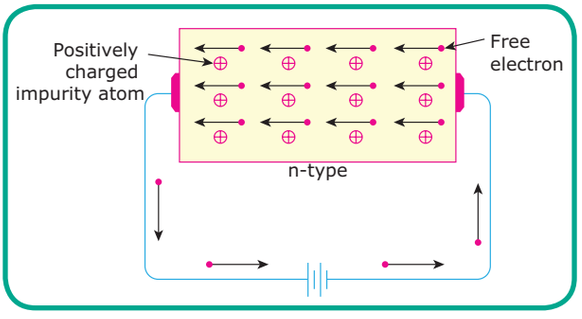


Figure 3.23 Current flow in n-type semiconductor

i. Many new free electrons are produced by the addition of pentavalent impurity.

ii. Thermal energy at room temperature still generates few hole-electron pairs.

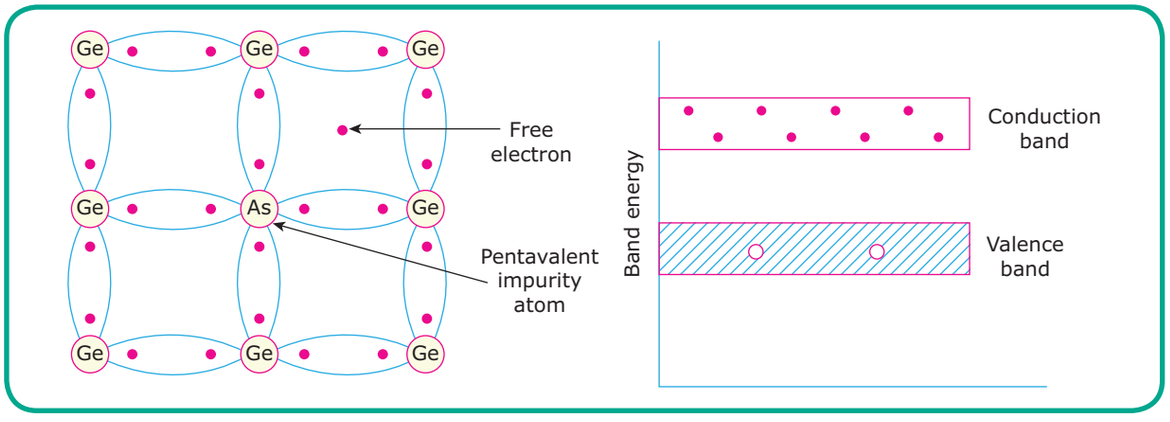


Figure 3.22 Doping of Ge with pentavalent impurity Atom As

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